

The Teukolsky–Starobinsky identities

1 Kerr-AdS black holes

1.1 The radial case

1.1.1 The radial ODE and its solutions

Fix parameters subextremal Kerr-AdS parameters (a, M, L) with $M > 0$, $L > 0$. Define r_+ to be the largest root of

$$\Delta := (r^2 + a^2) \left(1 + \frac{r^2}{L^2} \right) - 2Mr. \quad (1.1)$$

We also set

$$\Xi := 1 - \frac{a^2}{L^2}. \quad (1.2)$$

For $\omega \in \mathbb{C}$ and $m \in \frac{1}{2}\mathbb{Z}$, it will be convenient to define

$$\xi := -i \frac{2Mr_+}{r_+ - r_-} (\omega - m\omega_+); \quad \omega_+ := \frac{a}{2Mr_+}. \quad (1.3)$$

We say (ω, m, λ) is an admissible frequency triple with respect to a and s if $(\omega, m, \lambda) \in \mathbb{R}^3$.

In this section, we discuss the radial ODE

$$\begin{aligned} \Delta^{-s} \frac{d}{dr} \left(\Delta^{1+s} \frac{d}{dr} \right) R_{m\lambda}^{[s],(a\omega)}(r) + \frac{1}{\Delta} \left[\Xi^2 (\omega(r^2 + a^2) - am)^2 - is\Xi (\omega(r^2 + a^2) - am) \frac{d\Delta}{dr} \right] R_{m\lambda}^{[s],(a\omega)}(r) \\ + \left[4is\omega\Xi r + \frac{2}{L^2} (1 + 3s + 2s^2)r^2 + s \left(1 + \frac{a^2}{L^2} \right) - \lambda + 2\Xi^2 am\omega - a^2\Xi^2\omega^2 \right] R_{m\lambda}^{[s],(a\omega)}(r) = 0, \end{aligned} \quad (1.4)$$

where $r \in (r_+, \infty)$.

The radial ODE (1.4) has regular singularities at every one of the four roots of Δ , and another regular singularity at $r = \infty$. By standard asymptotic analysis, we can span the solution space by two linearly independent asymptotic solutions at each of the singularities $r = r_+, \infty$. Our basis is given by the following:

Definition 1.1. Fix $M > 0$, $|a| \leq M$, $s \in \frac{1}{2}\mathbb{Z}$ and an admissible frequency triple (ω, m, λ) with respect to a and s .

1. Define $R_{\mathcal{H}^+}^{[s]}$ and $R_{\mathcal{H}^-}^{[s]}$ to be the unique classical solutions to the homogeneous radial ODE (1.4) as follows.

- (a) If $\omega - m\omega_+ \neq 0$,

- i. $R_{\mathcal{H}^+}^{[s]}(r)(r - r_+)^{-\xi+s}$ and $R_{\mathcal{H}^-}^{[s]}(r)(r - r_+)^{\xi}$ are both smooth at $r = r_+$,
- ii. $\left((r - r_+)^{-\xi+s} R_{\mathcal{H}^+}^{[s]} \right) \big|_{r=r_+} = 1$, and $\left((r - r_+)^{\xi} R_{\mathcal{H}^-}^{[s]} \right) \big|_{r=r_+} = 1$;

- (b) If $\omega = m\omega_+$, and $s \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}_{>0}$,

- i. $R_{\mathcal{H}^+}^{[s]}(r)(r - r_+)^s$ is smooth at $r = r_+$,
- ii. $\left((r - r_+)^s R_{\mathcal{H}^+}^{[s]} \right) \big|_{r=r_+} = 1$,

- (c) If $\omega = m\omega_+$, and $s \in \mathbb{Z}_{>0}$,

- i. $R_{\mathcal{H}^+}^{[s]}(r)$ is smooth at $r = r_+$,
- ii. $\left(R_{\mathcal{H}^+}^{[s]} \right) \big|_{r=r_+} = 1$,

2. Define $R_{\mathcal{I},1}^{[s]}$ and $R_{\mathcal{I},2}^{[s]}$ to be the unique classical solutions to the homogeneous radial ODE (1.4) with boundary conditions

- (a) $R_{\mathcal{I},1}^{[s]} r^{1+2s}$ and $R_{\mathcal{I},2}^{[s]} r^{2+2s}$ are smooth as functions of r^{-1} as $r \rightarrow \infty$,
- (b) $(R_{\mathcal{I},1}^{[s]} r^{1+2s})|_{r=\infty} = 1$, and $(R_{\mathcal{I},2}^{[s]} r^{2+2s})|_{r=\infty} = 1$,
- (c) $r^{1+2s} R_{\mathcal{I},1}^{[s]} - 1 = O(r^{-2})$ and $r^{2+2s} R_{\mathcal{I},2}^{[s]} - 1 = O(r^{-2})$ as $r \rightarrow \infty$.

Remark 1.2. The condition $r^{2+2s} R_{\mathcal{I},2}^{[s]} - 1 = O(r^{-2})$ follows from the previous constraints on $R_{\mathcal{I},2}^{[s]}$, through the recurrence relation for the Frobenius representation of this solution. On the contrary, the condition $r^{1+2s} R_{\mathcal{I},1}^{[s]} - 1 = O(r^{-2})$ is a choice that is made so that $R_{\mathcal{I},1}^{[s]}$ is uniquely specified.

1.1.2 The radial Teukolsky–Starobinsky identities

In this section, we will introduce the Teukolsky–Starobinsky identities. We begin by defining the differential operators

$$D_n^\pm = \frac{d}{dr} \pm i \left(\frac{\Xi \omega (r^2 + a^2)}{\Delta} - \frac{a \Xi m}{\Delta} \right) + \frac{n}{\Delta} \frac{d\Delta}{dr}. \quad (1.5)$$

The radial Teukolsky–Starobinsky identities are differential identities that relate solutions of the radial ODE (2.4) with spin $+s$ and spin $-s$. We will present a rigorous statement:

Proposition 1.3 (Radial Teukolsky–Starobinsky identities). *Fix subextremal (a, M, L) Kerr-AdS parameters and $s \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. Let (ω, m, λ) be an admissible frequency triple with respect to s and a . There are constants $\mathfrak{C}_{ji}^{[\pm s]} \in \mathbb{C}$, for $i, j \in \{1, 2\}$, which depend only on s , (a, M, L) and (ω, m, λ) , such that*

$$\begin{aligned} \Delta^s (D_0^+)^{2s} \left(\Delta^s R_{\mathcal{I},j}^{[+s]} \right) &= \mathfrak{C}_{j1}^{[+s]} R_{\mathcal{I},1}^{[-s]} + \mathfrak{C}_{j2}^{[+s]} R_{\mathcal{I},2}^{[-s]} \\ \Delta^s (D_0^-)^{2s} \left(R_{\mathcal{I},j}^{[-s]} \right) &= \mathfrak{C}_{j1}^{[-s]} \Delta^s R_{\mathcal{I},1}^{[+s]} + \mathfrak{C}_{j2}^{[-s]} \Delta^s R_{\mathcal{I},2}^{[+s]} \end{aligned} \quad (1.6)$$

for $j = 1, 2$. Furthermore, we have **at least** for $s \leq 5/2$ [to check other cases]

$$\begin{aligned} [\mathfrak{C}^{[0]}] &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\ [\mathfrak{C}^{[\pm s]}] &= (\pm 1)^{\sin^2(\pi s)} L^{\mp s} \begin{pmatrix} i^{\sin^2(\pi s)} (\mathfrak{p}_{s,0} \mp i \mathfrak{q}_s) & \pm i^{\cos^2(\pi s)} L \mathfrak{p}_{s,2} \\ \pm i^{\cos^2(\pi s)} L^{-1} \mathfrak{p}_{s,1} & i^{\sin^2(\pi s)} (\mathfrak{p}_{s,0} \pm i \mathfrak{q}_s) \end{pmatrix}, \quad s \neq 0. \end{aligned}$$

Here, $\hat{\Lambda} = \lambda + a^2 \Xi^2 \omega^2 - 2a \Xi^2 m \omega + |s| \left(1 + \frac{a^2}{L^2} \right)$, and $\mathfrak{q}_s, \mathfrak{p}_{s,0}, \mathfrak{p}_{s,1}, \mathfrak{p}_{s,2}$ are given by

- for $s = 1/2$,

$$\mathfrak{q}_{\frac{1}{2}} = 0, \quad \mathfrak{p}_{\frac{1}{2},0} = -L \Xi \omega, \quad \mathfrak{p}_{\frac{1}{2},1} = 1, \quad \mathfrak{p}_{\frac{1}{2},2} = \hat{\Lambda} - L^2 \Xi^2 \omega^2,$$

with $\mathfrak{p}_{\frac{1}{2},0}^2 + \mathfrak{p}_{\frac{1}{2},1} \mathfrak{p}_{\frac{1}{2},2} = \hat{\Lambda}$;

- for $s = 1$,

$$\begin{aligned} \mathfrak{q}_1 &= 0, \quad \mathfrak{p}_{1,0} = \hat{\Lambda} - 2L^2 \Xi^2 \omega^2, \quad \mathfrak{p}_{1,1} = -2L \Xi \omega, \\ \mathfrak{p}_{1,2} &= 2L^3 \Xi^3 \omega^3 - 2L \hat{\Lambda} \Xi \omega + \frac{2a}{L} (a \Xi \omega - \Xi m), \end{aligned}$$

with $\mathfrak{p}_{1,0}^2 + \mathfrak{p}_{1,1} \mathfrak{p}_{1,2} = \hat{\Lambda}^2 + 4a \Xi^2 m \omega - 4a^2 \Xi^2 \omega^2$;

- for $s = 3/2$,

$$\begin{aligned}
\mathfrak{q}_{\frac{3}{2}} &= \frac{2M}{L}, \\
\mathfrak{p}_{\frac{3}{2},0} &= -\frac{4a}{L}\Xi m + 3\frac{a^2}{L}\Xi\omega + L\Xi\omega(-1 - 3\hat{\Lambda} + 4L^2\Xi^2\omega^2), \\
\mathfrak{p}_{\frac{3}{2},1} &= 1 + \hat{\Lambda} - 4L^2\Xi^2\omega^2 + \frac{a^2}{L^2}, \\
\mathfrak{p}_{\frac{3}{2},2} &= \hat{\Lambda}^2 - 8a\Xi^2m\omega - L^2(1 + 5\hat{\Lambda})\Xi^2\omega^2 + 4L^4\Xi^4\omega^4 - \frac{a^2}{L^2}(4 - 7L^2\Xi^2\omega^2),
\end{aligned}$$

with

$$\begin{aligned}
\mathfrak{p}_{\frac{3}{2},0}^2 + \mathfrak{p}_{\frac{3}{2},1}\mathfrak{p}_{\frac{3}{2},2} &= \hat{\Lambda}^2 \left(\hat{\Lambda} + 1 + \frac{a^2}{L^2} \right) - 4\hat{\Lambda} \left(4a^2\Xi^2\omega^2 - 4a\Xi^2m\omega + \frac{a^2}{L^2} \right) \\
&\quad + 16a^2\Xi^2\omega^2 \left(1 + \frac{a^2}{L^2} \right) - \frac{4a^2}{L^2} \left(1 + \frac{a^2}{L^2} - 4\Xi^2m^2 + 8a\Xi^2m\omega \right);
\end{aligned}$$

- for $s = 2$,

$$\begin{aligned}
\mathfrak{q}_2 &= 12M\Xi\omega, \\
\mathfrak{p}_{2,0} &= 2\hat{\Lambda} + \hat{\Lambda}^2 - 20a\Xi^2m\omega - 8L^2(1 + \hat{\Lambda})\Xi^2\omega^2 + 8L^4\Xi^4\omega^4 + 2a^2 \left(\frac{\hat{\Lambda} - 6}{L^2} + 6\Xi^2\omega^2 \right), \\
\mathfrak{p}_{2,1} &= -8\frac{a}{L}\Xi m - 4L(2 + \hat{\Lambda})\Xi\omega + 8L^3\Xi^3\omega^3, \\
\mathfrak{p}_{2,2} &= -4\hat{\Lambda}(1 + \hat{\Lambda})L\Xi\omega + 4L^3(2 + 3\hat{\Lambda})\Xi^3\omega^3 - 8L^5\Xi^5\omega^5 - \frac{12a}{L}\hat{\Lambda}\Xi m + 32aL\Xi^3m\omega^2 \\
&\quad + \frac{8a^2}{L}(3 + \hat{\Lambda})\Xi\omega - 24a^2L\Xi^3\omega^3,
\end{aligned}$$

with

$$\begin{aligned}
\mathfrak{p}_{2,0}^2 + \mathfrak{p}_{2,1}\mathfrak{p}_{2,2} &= \hat{\Lambda}^2 \left(\hat{\Lambda} + 2 \left(1 + \frac{a^2}{L^2} \right) \right)^2 - 8\hat{\Lambda}^2 \left(5a^2\Xi^2\omega^2 - 5a\Xi^2m\omega + 3\frac{a^2}{L^2} \right) \\
&\quad + 48\hat{\Lambda} \left(\left(1 + \frac{a^2}{L^2} \right) a^2\Xi^2\omega^2 + a\Xi^2m\omega \left(1 - 3\frac{a^2}{L^2} \right) + 2\frac{a^2}{L^2}\Xi^2m^2 \right) \\
&\quad - 48\frac{a^2}{L^2}\hat{\Lambda} \left(1 + \frac{a^2}{L^2} \right) + 144 \left(a\Xi^2m\omega - a^2\Xi^2\omega^2 + \frac{a^2}{L^2} \right)^2;
\end{aligned}$$

- for $s = 5/2$,

$$\begin{aligned}
\mathfrak{q}_{\frac{5}{2}} &= \frac{6M}{L^3} \left(6a^2 + L^2(6 + \hat{\Lambda} - 8L^2\Xi^2\omega^2) \right), \\
\mathfrak{p}_{\frac{5}{2},0} &= -24\frac{a^3}{L^3}\Xi m + 15\frac{a^4}{L^3}\Xi\omega + 4\frac{a}{L}\Xi m(-6 - 5\hat{\Lambda} + 20L^2\Xi^2\omega^2) \\
&\quad + \frac{a^2}{L}(90\Xi\omega - 40L^2\Xi^3\omega^3) \\
&\quad - L\Xi\omega \left(9 + 5\hat{\Lambda}^2 - 40L^2\Xi^2\omega^2 + 16L^4\Xi^4\omega^4 - 20\hat{\Lambda}(-1 + L^2\Xi^2\omega^2) \right), \\
\mathfrak{p}_{\frac{5}{2},1} &= \frac{9a^4}{L^4} - 48a\Xi^2m\omega + 2\frac{a^2}{L^2}(-9 + 3\hat{\Lambda} + 4L^2\Xi^2\omega^2) \\
&\quad + \left(9 + \hat{\Lambda}^2 - 40L^2\Xi^2\omega^2 + 16L^4\Xi^4\omega^4 + \hat{\Lambda}(6 - 12L^2\Xi^2\omega^2) \right), \\
\mathfrak{p}_{\frac{5}{2},2} &= -\frac{144M^2}{L^2} + \hat{\Lambda}^2(4 + \hat{\Lambda}) + 80\frac{a^3}{L^2}\Xi^2m\omega - 25\frac{a^4}{L^2}\Xi^2\omega^2 \\
&\quad - L^2(9 + 34\hat{\Lambda} + 13\hat{\Lambda}^2)\Xi^2\omega^2 + 4L^4(10 + 7\hat{\Lambda})\Xi^4\omega^4 - 16L^6\Xi^6\omega^6 \\
&\quad + 8a\Xi^2m\omega(-6 - 9\hat{\Lambda} + 14L^2\Xi^2\omega^2) \\
&\quad + \frac{a^2}{L^2} \left(4\hat{\Lambda}^2 - 64\Xi^2m^2 + 162L^2\Xi^2\omega^2 - 72L^4\Xi^4\omega^4 - \hat{\Lambda}(48 - 38L^2\Xi^2\omega^2) \right).
\end{aligned}$$

with $\mathfrak{p}_{\frac{5}{2},0}^2 + \mathfrak{p}_{\frac{5}{2},1}\mathfrak{p}_{\frac{5}{2},2}$ **NOT** equal to the angular TS constant and hence possibly not positive [?].

In order to prove Proposition 1.3, we quote the following two results, obtained from adapting [KMW89] to the asymptotically AdS case:

Lemma 1.4. *For solutions of the homogeneous radial ODE (1.4) with spin $\pm s$, we set*

$$P^{[+s]} := \Delta^s R^{[+s]}, \quad P^{[-s]} := R^{[-s]}.$$

Then, we have

$$\begin{aligned}
&\left[\Delta D_{1-s}^\mp D_0^\pm \pm 2(2s-1)i\Xi\omega r + \frac{2r^2}{L^2}(1-3|s|+2s^2) \right] P^{[\pm s]}(r) \\
&= \left(\lambda + |s| \left(1 + \frac{a^2}{L^2} \right) + a^2\Xi^2\omega^2 - 2a\Xi^2m\omega \right) P^{[\pm s]}(r).
\end{aligned} \tag{1.7}$$

By induction on $2s$, one can also show that:

Lemma 1.5. *The operators $D_0^\pm$, defined in (1.5), satisfy the commutation relation*

$$\begin{aligned}
\Delta^s (D_0^\mp)^{2s} &\left[\Delta D_{1-s}^\pm D_0^\mp \mp 2(2s-1)i\Xi\omega r + \frac{2r^2}{L^2}(1-3|s|+2s^2) \right] \\
&= \left[\Delta D_{1-s}^\mp D_0^\pm \pm 2(2s-1)i\Xi\omega r + \frac{2r^2}{L^2}(1-3|s|+2s^2) \right] \Delta^s (D_0^\mp)^{2s}. \tag{1.8}
\end{aligned}$$

Combining the two lemmas, we see that, if $P^{[\pm s]}$ solves (2.9) with spin $\pm$, then $\Delta^s (D_0^\pm)^{2s} P^{[\pm s]}$ is a solution of the homogeneous radial ODE (2.9) for spin $\mp s$.

Proof of Proposition 2.3. The statement is trivial for $s = 0$. For $s \neq 0$, we obtain the result by considering the Frobenius representations of $R_{\mathcal{I},1}^{[\pm s]}$ and $R_{\mathcal{I},2}^{[\pm s]}$, on which we can act directly with

the Teukolsky–Starobinsky operators. In terms of $x = 1/r$, these are

$$R_{\mathcal{I},1}^{[\pm s]} = x^{1 \pm 2s} \left(1 + \sum_{k=2}^{\infty} c_{k,1}^{[\pm s]} x^k \right),$$

$$R_{\mathcal{I},2}^{[\pm s]} = x^{2 \pm 2s} \left(1 + \sum_{k=2}^{\infty} c_{k,2}^{[\pm s]} x^k \right).$$

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1.1.3 An energy current

In this section, we derive an energy current for the radial ODE (1.4). To do, it is convenient to recast it in terms of another variable: writing $u_{m\lambda}^{[s],(a\omega)} = (r^2 + a^2)^{1/2} \Delta^{s/2} R_{m\lambda}^{[s],(a\omega)}$, we have

$$\begin{aligned} \partial_{r^*}^2 u_{m\lambda}^{[s],(a\omega)} + (\Xi^2 \omega^2 - V_{m\lambda}^{[s],(a\omega)}) u_{m\lambda}^{[s],(a\omega)} &= 0, \\ V_{m\lambda}^{[s],(a\omega)} &= \frac{2am\Xi^2\omega}{r^2 + a^2} - \frac{a^2\Xi^2m^2}{(r^2 + a^2)^2} + is \frac{\partial_r \Delta}{r^2 + a^2} \left(\Xi\omega - \frac{a\Xi m}{(r^2 + a^2)} \right) \\ &\quad - \frac{\Delta}{(r^2 + a^2)^2} \left(\frac{2r^2}{L^2} - \lambda + 2\Xi^2 am\omega - a^2\Xi^2\omega^2 + 4is\Xi\omega r \right) + \partial_{r^*} \left(\frac{r\Delta}{(r^2 + a^2)^2} \right) \\ &\quad + s^2 \left[\frac{(\partial_r \Delta)^2}{4(r^2 + a^2)^2} - \frac{4r^2\Delta}{L^2(r^2 + a^2)^2} \right] + \left(\frac{r\Delta}{(r^2 + a^2)^2} \right)^2, \end{aligned} \tag{1.9}$$

where we have introduced a coordinate r^* : $(-\infty, \pi/2)$ defined via

$$\frac{dr^*}{dr} = \frac{r^2 + a^2}{\Delta}, \quad r^*(r) \rightarrow \pi/2 \text{ as } r \rightarrow \infty.$$

Our energy current is based on the following conservation law:

Lemma 1.6. *Fix (a, M, L) subextremal and $s \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. If (ω, m, λ) is an admissible frequency triple with respect to s and a , then*

$$\frac{d}{dr^*} W[\overline{u^{[+s]}}, u^{[-s]}] = 0, \quad W[\overline{u^{[+s]}}, u^{[-s]}] = \overline{u^{[+s]}} \frac{d}{dr^*} u^{[-s]} - u^{[-s]} \frac{d}{dr^*} \overline{u^{[+s]}}$$

We then have

Proposition 1.7 (Energy current). *[To FILL]*

Proof. Note that there are constants $a_{\mathcal{I},1}^{[\pm s]}, a_{\mathcal{I},2}^{[\pm s]}$ such that

$$u^{[\pm s]} = a_{\mathcal{I},1}^{[\pm s]} \Delta^{\pm s/2} (r^2 + a^2)^{1/2} R_{\mathcal{I},1}^{[\pm s]} + a_{\mathcal{I},2}^{[\pm s]} \Delta^{\pm s/2} (r^2 + a^2)^{1/2} R_{\mathcal{I},2}^{[\pm s]},$$

so we have

$$\lim_{r \rightarrow \infty} W[\overline{u^{[+s]}}, u^{[-s]}](r^*(r)) = \frac{1}{L^2} (a_{\mathcal{I},1}^{[-s]}, \overline{a_{\mathcal{I},2}^{[+s]}} - a_{\mathcal{I},2}^{[-s]}, \overline{a_{\mathcal{I},1}^{[+s]}}).$$

To obtain a conservation law for $\pm s$, we apply Lemma 1.6 with $u^{[\mp s]}$ obtained from $u^{[\pm s]}$ via the Teukolsky–Starobinsky map in Proposition 1.3. In this case, $a_{\mathcal{I},1}^{[\pm s]}$ and $a_{\mathcal{I},2}^{[\pm s]}$ are free but $a_{\mathcal{I},1}^{[\mp s]}$ and $a_{\mathcal{I},2}^{[\mp s]}$ are determined by the aforementioned proposition:

$$\begin{aligned} a_{\mathcal{I},1}^{[\mp s]} &= \mathfrak{C}_{11}^{[\pm s]} a_{\mathcal{I},1}^{[\pm s]} + \mathfrak{C}_{21}^{[\pm s]} a_{\mathcal{I},2}^{[\pm s]} \\ a_{\mathcal{I},2}^{[\mp s]} &= \mathfrak{C}_{12}^{[\pm s]} a_{\mathcal{I},1}^{[\pm s]} + \mathfrak{C}_{22}^{[\pm s]} a_{\mathcal{I},2}^{[\pm s]} \end{aligned}$$

Then, we have

$$\begin{aligned} \lim_{r \rightarrow \infty} W[\overline{u^{[+s]}}, u^{[-s]}](r^*(r)) &= \frac{1}{L^2} \left(-\mathfrak{C}_{12}^{[+s]} |a_{\mathcal{I},1}^{[+s]}|^2 + \mathfrak{C}_{21}^{[+s]} |a_{\mathcal{I},2}^{[+s]}|^2 \right. \\ &\quad \left. + \mathfrak{C}_{11}^{[+s]} a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}} - \mathfrak{C}_{22}^{[+s]} a_{\mathcal{I},2}^{[+s]} \overline{a_{\mathcal{I},1}^{[+s]}} \right), \end{aligned}$$

and

$$\begin{aligned} \lim_{r \rightarrow \infty} W[\overline{u^{[+s]}}, u^{[-s]}](r^*(r)) &= \frac{1}{L^2} \left(\overline{\mathfrak{C}_{12}^{[-s]}} |a_{\mathcal{I},1}^{[-s]}|^2 - \overline{\mathfrak{C}_{21}^{[-s]}} |a_{\mathcal{I},2}^{[-s]}|^2 \right. \\ &\quad \left. + \overline{\mathfrak{C}_{22}^{[-s]}} a_{\mathcal{I},1}^{[-s]} \overline{a_{\mathcal{I},2}^{[-s]}} - \overline{\mathfrak{C}_{11}^{[-s]}} a_{\mathcal{I},2}^{[-s]} \overline{a_{\mathcal{I},1}^{[-s]}} \right), \end{aligned}$$

respectively. For instance, using Proposition 1.3, we have

$$\begin{aligned} \lim_{r \rightarrow \infty} W[\overline{u^{[+s]}}, u^{[-s]}](r^*(r)) &= i^{\cos^2(\pi s)} L^{-(2s+2)} \left(-(-1)^{2s} L \mathfrak{p}_{s,2} |a_{\mathcal{I},1}^{[+s]}|^2 + (-1)^{2s} L^{-1} \mathfrak{p}_{s,1} |a_{\mathcal{I},2}^{[+s]}|^2 \right. \\ &\quad \left. + 2\mathfrak{p}_{s,0} \Im(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}}) - 2\mathfrak{q}_s \Re(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}}) \right) \end{aligned}$$

in the first case, and [COMPLETE ME WITH SECOND CASE] □

Lemma 1.8. Suppose $\mathfrak{P}_s := \mathfrak{p}_{s,0}^2 + \mathfrak{p}_{s,1}\mathfrak{p}_{s,2} > 0$. Then,

$$\begin{pmatrix} \mathfrak{p}_{s,0} \pm \sqrt{\mathfrak{P}_s} & i(-1)^{2s} L^{-1} \mathfrak{p}_{s,1} \\ i(-1)^{2s} L \mathfrak{p}_{s,2} & \mathfrak{p}_{s,0} \mp \sqrt{\mathfrak{P}_s} \end{pmatrix} \begin{pmatrix} a_{\mathcal{I},1}^{[+s]} \\ a_{\mathcal{I},2}^{[+s]} \end{pmatrix} = 0 \implies \lim_{r \rightarrow \infty} W[\overline{u^{[+s]}}, u^{[-s]}](r^*(r)) = 0.$$

The converse is not true.

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Proof. A simple computation yields

$$\begin{aligned} &\begin{pmatrix} \mathfrak{p}_{s,0} \pm \sqrt{\mathfrak{P}_s} & i(-1)^{2s} L^{-1} \mathfrak{p}_{s,1} \\ i(-1)^{2s} L \mathfrak{p}_{s,2} & \mathfrak{p}_{s,0} \mp \sqrt{\mathfrak{P}_s} \end{pmatrix} \begin{pmatrix} a_{\mathcal{I},1}^{[+s]} \\ a_{\mathcal{I},2}^{[+s]} \end{pmatrix} = 0 \\ \implies &\begin{cases} (-1)^{2s} L^{-1} \mathfrak{p}_{s,1} |a_{\mathcal{I},2}^{[+s]}|^2 = i(\mathfrak{p}_{s,0} \pm \sqrt{\mathfrak{P}_s}) a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}} \\ (-1)^{2s} L^{-1} \mathfrak{p}_{s,2} |a_{\mathcal{I},1}^{[+s]}|^2 = i(\mathfrak{p}_{s,0} \mp \sqrt{\mathfrak{P}_s}) a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}} \end{cases}. \end{aligned}$$

Subtracting the two equations, we get

$$-(-1)^{2s} L \mathfrak{p}_{s,2} |a_{\mathcal{I},1}^{[+s]}|^2 + (-1)^{2s} L^{-1} \mathfrak{p}_{s,1} |a_{\mathcal{I},2}^{[+s]}|^2 = -2\mathfrak{p}_{s,0} \Im(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}}) + 2i\sqrt{\mathfrak{P}_s} \Re(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}});$$

on the other hand, taking imaginary parts of the two equations, we obtain

$$(\mathfrak{p}_{s,0} \mp \sqrt{\mathfrak{P}_s}) \Re(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}}) = 0 = (\mathfrak{p}_{s,0} \pm \sqrt{\mathfrak{P}_s}) \Re(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}}) \implies \Re(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}}) = 0,$$

since $\sqrt{\mathfrak{P}_s} \neq 0$. Thus, we have

$$\lim_{r \rightarrow \infty} W[\overline{u^{[+s]}}, u^{[-s]}](r^*(r)) = -2i^{\cos^2(\pi s)} L^{-(2s+2)} \mathfrak{q}_s \Re(a_{\mathcal{I},1}^{[+s]} \overline{a_{\mathcal{I},2}^{[+s]}}) = 0.$$

□

1.2 Comparison with Graf–Holzegel

In comparing the section above with the paper of Graf–Holzegel, it is important to note that

$$\begin{aligned} L &= 1/k_{\text{GH}}, \\ \hat{\Lambda} &= \lambda_{m\ell, \text{GH}}^\omega, \\ R_{m\lambda}^{[\pm 2], (a\omega)} &= \frac{(r^2 + a^2)^{3/2}}{\Delta_{(1\pm 1)}} R_{[\pm 2], \omega, \text{GH}}^{m\ell}, \end{aligned}$$

where the subscript GH refers to the Graf–Holzegel variables.

The Robin boundary conditions of Graf–Holzegel’s appendix are, in our notation,

$$\mathcal{R}_\pm \begin{pmatrix} a_{\mathcal{I},1}^{[+2]} \\ a_{\mathcal{I},2}^{[+2]} \end{pmatrix} = 0, \quad \mathcal{R}_\pm \begin{pmatrix} a_{\mathcal{I},1}^{[-2]} \\ a_{\mathcal{I},2}^{[-2]} \end{pmatrix} = 0,$$

where

$$\mathcal{R}_\pm := \begin{pmatrix} \mathfrak{p}_{2,0} \pm \sqrt{\mathfrak{p}_{2,0}^2 + \mathfrak{p}_{2,1}\mathfrak{p}_{2,2}} & iL^{-1}\mathfrak{p}_{2,1} \\ iL\mathfrak{p}_{2,2} & \mathfrak{p}_{2,0} \mp \sqrt{\mathfrak{p}_{2,0}^2 + \mathfrak{p}_{2,1}\mathfrak{p}_{2,2}} \end{pmatrix}.$$

1.3 Comparison with Wang–Herdeiro–Jing/Sampaio

In comparing the section above with the paper of Wang–Herdeiro–Sampaio, it is important to note that (comparing the radial ODEs)

$$\begin{aligned} \hat{\Lambda} &= \lambda_{\text{WHS}} - \frac{a^2}{L^2} \\ R_{m\lambda}^{[\pm 1], (a\omega)} &= R_{\pm 1, \text{WHS}}, \end{aligned}$$

where the subscript WHS refers to the Wang–Herdeiro–Sampaio variables.

The Robin boundary conditions of Wang–Herdeiro–Sampaio appendix are, in our notation, [FILL ME].

[ADD COMPARISON TO Wang–Herdeiro–Jing]

2 Kerr black holes

2.1 The radial case

2.1.1 The radial ODE and its solutions

Fix parameters (a, M) with $M > 0$ and $|a| \leq M$ and define

$$r_\pm := M \pm \sqrt{M^2 - a^2}. \quad (2.1)$$

We write

$$\Delta := r^2 - 2Mr + a^2 = (r - r_+)(r - r_-). \quad (2.2)$$

For $\omega \in \mathbb{C}$ and $m \in \frac{1}{2}\mathbb{Z}$, it will be convenient to define

$$\xi := -i \frac{2Mr_+}{r_+ - r_-} (\omega - m\omega_+), \quad \text{if } |a| < M; \quad \beta := 2iM^2(\omega - m\omega_+); \quad \omega_+ := \frac{a}{2Mr_+}. \quad (2.3)$$

We say (ω, m, λ) is an admissible frequency triple with respect to a and s if $(\omega, m, \lambda) \in \mathbb{R}^3$, $\omega \neq 0$ and, for $|a| = M$, if $\omega \neq m\omega_+$ too.

In this section, we discuss the radial ODE

$$\Delta^{-s} \frac{d}{dr} \left(\Delta^{s+1} \frac{d}{dr} \right) R_{m\lambda}^{[s],(a\omega)}(r) + \left(\frac{[\omega(r^2 + a^2) - am]^2 - 2is(r - M)[\omega(r^2 + a^2) - am]}{\Delta} \right) R_{m\lambda}^{[s],(a\omega)}(r) + (4is\omega r - \lambda + s - a^2\omega^2 + 2am\omega) R_{m\lambda}^{[s],(a\omega)}(r) = 0 \quad (2.4)$$

where $r \in (r_+, \infty)$.

The radial ODE (2.4) has singularities at $r = r_{\pm}$, which are regular if $|a| < M$ but are a single irregular singularity of rank 1 if $|a| = M$, and an irregular singularity of rank 1 at $r = \infty$. By standard asymptotic analysis, we can span the solution space by two linearly independent asymptotic solutions at each of the singularities $r = r_+, \infty$. Our basis is given by the following:

Definition 2.1. Fix $M > 0$, $|a| \leq M$, $s \in \frac{1}{2}\mathbb{Z}$ and an admissible frequency triple (ω, m, λ) with respect to a and s .

1. Define $R_{\mathcal{H}+}^{[s]}$ and $R_{\mathcal{H}-}^{[s]}$ to be the unique classical solutions to the homogeneous radial ODE (2.4) with boundary conditions
 - (a) if $|a| < M$ and additionally $\omega - m\omega_+ \neq 0$,
 - i. $R_{\mathcal{H}+}^{[s]}(r)(r - r_+)^{-\xi+s}$ and $R_{\mathcal{H}-}^{[s]}(r)(r - r_+)^{\xi}$ are both smooth at $r = r_+$,
 - ii. $\left((r - r_+)^{-\xi+s} R_{\mathcal{H}+}^{[s]}\right)|_{r=r_+} = 1$, and $\left((r - r_+)^{\xi} R_{\mathcal{H}-}^{[s]}\right)|_{r=r_+} = 1$;
 - (b) if $|a| < M$, $\omega = m\omega_+$, and $s \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}_{>0}$,
 - i. $R_{\mathcal{H}+}^{[s]}(r)(r - r_+)^s$ is smooth at $r = r_+$,
 - ii. $\left((r - r_+)^s R_{\mathcal{H}+}^{[s]}\right)|_{r=r_+} = 1$,
 - (c) if $|a| < M$, $\omega = m\omega_+$, and $s \in \mathbb{Z}_{>0}$,
 - i. $R_{\mathcal{H}+}^{[s]}(r)$ is smooth at $r = r_+$,
 - ii. $\left(R_{\mathcal{H}+}^{[s]}\right)|_{r=r_+} = 1$,
 - (d) if $|a| = M$ and additionally $\omega - m\omega_+ \neq 0$,
 - i. $(r - M)^{2iM\omega+2s} R_{\mathcal{H}+}^{[s]}(r)e^{-\beta(r-M)^{-1}}$ and $(r - M)^{-2iM\omega} R_{\mathcal{H}-}^{[s]}(r)e^{\beta(r-M)^{-1}}$ are both smooth at $r = M$,
 - ii. $\left((r - M)^{2iM\omega+2s} e^{-\beta(r-M)^{-1}} R_{\mathcal{H}+}^{[s]}\right)|_{r=M} = 1$, and $\left((r - M)^{-2iM\omega} e^{\beta(r-M)^{-1}} R_{\mathcal{H}-}^{[s]}\right)|_{r=M} = 1$.
2. Define $R_{\mathcal{I}+}^{[s]}$ and $R_{\mathcal{I}-}^{[s]}$ to be the unique classical solutions to the homogeneous radial ODE (2.4) and boundary conditions
 - (a) $R_{\mathcal{I}+}^{[s]} \sim e^{i\omega r} r^{2Mi\omega-2s-1}$ and $R_{\mathcal{I}-}^{[s]} \sim e^{-i\omega r} r^{-2Mi\omega-1}$ asymptotically¹ as $r \rightarrow \infty$,
 - (b) $\left(e^{-i\omega r} r^{-2iM\omega+1+2s} R_{\mathcal{I}+}^{[s]}\right)|_{r=\infty} = 1$, and $\left(e^{i\omega r} r^{2iM\omega+1} R_{\mathcal{I}-}^{[s]}\right)|_{r=\infty} = 1$.

In light of the previous comments, we find that we have the following representation for solutions of (2.4):

Lemma 2.2. Fix $M > 0$, $|a| \leq M$, $s \in \frac{1}{2}\mathbb{Z}$ and an admissible frequency triple (ω, m, λ) with respect to a and s such that $\omega \neq m\omega_+$. A solution $R_{m\lambda}^{[s],(a\omega)}$ to the homogeneous radial ODE (2.4) can be written as, dropping most sub and superscripts,

$$R^{[s]} = a_{\mathcal{H}+}^{[s]} R_{\mathcal{H}+}^{[s]} + a_{\mathcal{H}-}^{[s]} R_{\mathcal{H}-}^{[s]}, \quad R^{[s]} = a_{\mathcal{I}+}^{[s]} R_{\mathcal{I}+}^{[s]} + a_{\mathcal{I}-}^{[s]} R_{\mathcal{I}-}^{[s]}, \quad (2.5)$$

for some $a_{\mathcal{H}\pm}^{[s]}, a_{\mathcal{I}\pm}^{[s]} \in \mathbb{C}$.

¹We write $f \sim g$ asymptotically as $r \rightarrow \infty$ to mean that for every $N \geq 0$, we have complex numbers $c_1, \dots, c_N$ such that $f(r) = g(r) \left[1 + \sum_{k=1}^N c_k r^{-k} + O(r^{-N-1})\right]$.

2.1.2 The radial Teukolsky–Starobinsky identities

In this section, we will introduce the Teukolsky–Starobinsky identities. We begin by defining the differential operators

$$D_n^\pm = \frac{d}{dr} \pm i \left(\frac{\omega(r^2 + a^2)}{\Delta} - \frac{am}{\Delta} \right) + \frac{2n(r - M)}{\Delta}. \quad (2.6)$$

The radial Teukolsky–Starobinsky identities are differential identities that relate solutions of the radial ODE (2.4) with spin $+s$ and spin $-s$, and were introduced in [TP74, SC74] and extended for general s in [KMW89]. We will present a rigorous statement:

Proposition 2.3 (Radial Teukolsky–Starobinsky identities). *Fix $M > 0$, $|a| \leq M$ and $s \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. Let (ω, m, λ) be an admissible frequency triple with respect to s and a . Dropping most subscripts, let $R^{[\pm s]}$ be solutions to the homogeneous radial ODE (2.4) of spin $\pm s$. If $\omega \neq m\omega_+$, consider the representation (2.5)*

$$\begin{aligned} R^{[\pm s]} &= a_{\mathcal{H}^+}^{[\pm s]} R_{\mathcal{H}^+}^{[\pm s]} + a_{\mathcal{H}^-}^{[\pm s]} R_{\mathcal{H}^-}^{[\pm s]} \\ &= a_{\mathcal{I}^+}^{[\pm s]} R_{\mathcal{I}^+}^{[\pm s]} + a_{\mathcal{I}^-}^{[\pm s]} R_{\mathcal{I}^-}^{[\pm s]}; \end{aligned}$$

otherwise, we consider outgoing boundary conditions, i.e.

$$\begin{aligned} R^{[\pm s]} &= a_{\mathcal{H}^+}^{[\pm s]} R_{\mathcal{H}^+}^{[\pm s]} \\ &= a_{\mathcal{I}^+}^{[\pm s]} R_{\mathcal{I}^+}^{[\pm s]}, \end{aligned}$$

and set $a_{\mathcal{H}^-}^{[\pm s]} = 0 = a_{\mathcal{I}^-}^{[\pm s]}$ by convention. Then we have

$$\begin{aligned} \Delta^s (D_0^+)^{2s} (\Delta^s R^{[+s]}) &= \mathfrak{C}_s^{(1)} a_{\mathcal{I}^+}^{[+s]} R_{\mathcal{I}^+}^{[-s]} + \mathfrak{C}_s^{(7)} a_{\mathcal{I}^-}^{[+s]} R_{\mathcal{I}^-}^{[-s]} \\ &= \mathfrak{C}_s^{(4)} a_{\mathcal{H}^+}^{[+s]} R_{\mathcal{H}^+}^{[-s]} + \mathfrak{C}_s^{(6)} a_{\mathcal{H}^-}^{[+s]} R_{\mathcal{H}^-}^{[-s]}, \\ \Delta^s (D_0^-)^{2s} R^{[-s]} &= \mathfrak{C}_s^{(3)} a_{\mathcal{I}^+}^{[-s]} \Delta^s R_{\mathcal{I}^+}^{[+s]} + \mathfrak{C}_s^{(5)} a_{\mathcal{I}^-}^{[-s]} \Delta^s R_{\mathcal{I}^-}^{[+s]} \\ &= \mathfrak{C}_s^{(2)} a_{\mathcal{H}^+}^{[-s]} \Delta^s R_{\mathcal{H}^+}^{[+s]} + \mathfrak{C}_s^{(8)} a_{\mathcal{H}^-}^{[-s]} \Delta^s R_{\mathcal{H}^-}^{[+s]}, \end{aligned} \quad (2.7)$$

for some $\mathfrak{C}_s^{(i)} \in \mathbb{C}$, $i = 1, \dots, 8$, which depend only on s , (a, M) and (ω, m, λ) and such that $\mathfrak{C}_0^{(i)} = 1$ for $s = 0$ and, for $s \neq 0$,

$$\mathfrak{C}_s^{(1)} = (2i\omega)^{2s}, \quad \mathfrak{C}_s^{(5)} = (-2i\omega)^{2s},$$

and

$$\begin{aligned} \mathfrak{C}_s^{(2)} &= \begin{cases} \prod_{j=0}^{2s-1} (2\xi + s - j), & |a| < M, \text{ and either } \omega \neq m\omega_+ \text{ or } s \in \frac{1}{2}\mathbb{Z}_{\geq 0} \setminus \mathbb{Z}, \\ (-2\beta)^{2s}, & |a| = M, \end{cases} \\ \mathfrak{C}_s^{(6)} &= \begin{cases} (r_+ - r_-)^{2s} \prod_{j=0}^{2s-1} (-2\xi + s - j), & |a| < M, \text{ and either } \omega \neq m\omega_+ \text{ or } s \in \frac{1}{2}\mathbb{Z}_{\geq 0} \setminus \mathbb{Z}, \\ (2\beta)^{2s}, & |a| = M. \end{cases} \end{aligned} \quad (2.8)$$

In order to prove Proposition 2.3, we recall the following two results from [KMW89]:

Lemma 2.4. *For solutions of the homogeneous radial ODE (2.4) with spin $\pm s$, we set*

$$P^{[+s]} := \Delta^s R^{[+s]}, \quad P^{[-s]} := R^{[-s]}.$$

Then, we have

$$\left[\Delta D_{1-s}^\mp D_0^\pm \pm 2(2s-1)i\omega r \right] P^{[\pm s]}(r) = \left(\lambda + |s| + a^2\omega^2 - 2am\omega \right) P^{[\pm s]}(r). \quad (2.9)$$

By induction on $2s$, one can also show that:

Lemma 2.5. *The operators $D_0^\pm$, defined in (2.6), satisfy the commutation relation*

$$\begin{aligned} \Delta^s (D_0^\mp)^{2s} \left[\Delta D_{1-s}^\pm D_0^\mp \mp 2(2s-1)i\omega r \right] \\ = \left[\Delta D_{1-s}^\mp D_0^\pm \pm 2(2s-1)i\omega r \right] \Delta^s (D_0^\mp)^{2s}. \end{aligned} \quad (2.10)$$

Combining the two lemmas, we see that, if $P^{[\pm s]}$ solves (2.9) with spin $\pm$, then $\Delta^s (D_0^\pm)^{2s} P^{[\pm s]}$ is a solution of the homogeneous radial ODE (2.9) for spin $\mp s$. We are now ready to prove Proposition 2.3:

Proof of Proposition 2.3. The statement is trivial for $s = 0$. For $s \neq 0$, we obtain the result by using the asymptotic representations of $R_{\mathcal{H}^+}^{[\pm s]}$, $R_{\mathcal{I}^+}^{[\pm s]}$, $R_{\mathcal{H}^-}^{[\pm s]}$ and $R_{\mathcal{I}^-}^{[\pm s]}$, on which we can act directly with the Teukolsky–Starobinsky operators. Since $\omega \neq 0$, these are

$$\begin{aligned} \Delta^{\frac{s \pm s}{2}} R_{\mathcal{I}^+}^{[\pm s]} &= e^{i\omega r} r^{2iM\omega + s \mp s - 1} \left[\sum_{k=0}^{2s} c_k^{[\pm s]} r^{-k} + O(r^{-2s-1}) \right], \\ \Delta^{\frac{s \pm s}{2}} R_{\mathcal{I}^-}^{[\pm s]} &= e^{-i\omega r} r^{-2iM\omega + s \pm s - 1} \left[\sum_{k=0}^{2s} c_{k,2}^{[\pm s]} r^{-k} + O(r^{-2s-1}) \right], \end{aligned} \quad (2.11)$$

as $r \rightarrow \infty$; if $\omega \neq m\omega_+$,

$$\begin{aligned} \Delta^{\frac{s \pm s}{2}} R_{\mathcal{H}^+}^{[\pm s]} &= \begin{cases} (r - r_-)^{\frac{s \pm s}{2}} \sum_{k=0}^{\infty} b_k^{[\pm s]} (r - r_+)^{\xi + (s \mp s)/2 + k}, & |a| < M, \\ \sum_{k=0}^{\infty} b_k^{[\pm s]} e^{\beta(r-M)^{-1}} (r - M)^{-2iM\omega + s \mp s + k}, & |a| = M, \end{cases} \\ \Delta^{\frac{s \pm s}{2}} R_{\mathcal{H}^-}^{[\pm s]} &= \begin{cases} (r - r_-)^{\frac{s \pm s}{2}} \sum_{k=0}^{\infty} b_{k,2}^{[\pm s]} (r - r_+)^{-\xi + (s \pm s)/2 + k}, & |a| < M, \\ \sum_{k=0}^{\infty} b_{k,2}^{[\pm s]} e^{-\beta(r-M)^{-1}} (r - M)^{2iM\omega + s \pm s + k}, & |a| = M, \end{cases} \end{aligned} \quad (2.12)$$

as $r \rightarrow r_+$; and, if $\omega = m\omega_+$,

$$\Delta^{\frac{s \pm s}{2}} R_{\mathcal{H}^+}^{[\pm s]} = (r - r_-)^{\frac{s \pm s}{2}} \sum_{k=0}^{\infty} b_k^{[\pm s]} (r - r_+)^{\xi + (s \mp s)/2 + k}, \quad |a| < M, \quad s \in \frac{1}{2}\mathbb{Z}_{\geq 0} \setminus \mathbb{Z} \quad (2.13)$$

$$R_{\mathcal{H}^+}^{[-s]} = \sum_{k=0}^{\infty} b_k^{[-s]} (r - r_+)^{\xi + s + k}, \quad |a| < M, \quad s \in \mathbb{Z}_{\geq 0}, \quad (2.14)$$

$$\Delta^s R_{\mathcal{H}^+}^{[+s]} = (r - r_-)^s \sum_{k=0}^{\infty} b_k^{[+s]} (r - r_+)^{\xi + s + k}, \quad |a| < M, \quad s \in \mathbb{Z}_{\geq 0}, \quad (2.15)$$

as $r \rightarrow r_+$. By the normalizations, $c_0^{[\pm s]} = c_{0,2}^{[\pm s]} = b_0^{[\pm s]} = b_{0,2}^{[\pm s]} = 1$.

For computations at $r = r_+$, it will be useful to recall from the definition of ξ and β in (2.3),

$$\begin{aligned} \xi(r - r_-) + i\omega(r^2 + a^2) - iam &= \xi(r_+ - r_-) + i\omega(r_+^2 + a^2) - iam \\ &\quad + \xi(r - r_+) + i\omega(r - r_+)(r + r_+) \\ &= \xi(r - r_+) + i\omega(r - r_+)(r + r_+), \end{aligned} \quad (2.16)$$

$$\beta + i\omega(r - M)(r + M) = i\omega(r^2 + M^2) - iam. \quad (2.17)$$

The cases $R_{\mathcal{I}^+}^{[+s]}$ and $R_{\mathcal{H}^+}^{[-s]}$. In these cases, the constant arising from application of the Teukolsky–Starobinsky operators to these functions involves only the first term of the series.

We begin at $r \rightarrow \infty$, with positive spin. For any $s \geq \frac{1}{2}$, we will have

$$\begin{aligned} D_0^+ (\Delta^s R_{\mathcal{I}^+}^{[+s]}) &= \left(\frac{d}{dr} + i \frac{\omega(r^2 + a^2) - am}{\Delta} \right) (\Delta^s R_{\mathcal{I}^+}^{[+s]}) \\ &= e^{i\omega r} r^{2iM\omega-1} \left[\sum_{k=0}^{2s} c_k^{[+s]} \left(2i\omega - \frac{k+1-2iM\omega}{r} + \frac{2Mi\omega r - iam}{\Delta} \right) r^{-k} + O(r^{-2s-1}) \right]. \end{aligned}$$

Repeating $2s-1$ times, we obtain

$$\Delta^s (D_0^+)^{2s} (\Delta^s R_{\mathcal{I}^+}^{[+s]}) = e^{i\omega r} r^{2iM\omega+2s-1} \left[\sum_{k=0}^{2s} c_k^{[+s]} \left((2i\omega)^{2s} + O(r^{-1}) \right) r^{-k} + O(r^{-2s-1}) \right]. \quad (2.18)$$

By Lemma 2.5, (2.18) satisfies the spin $-s$ radial ODE, hence, by Definition 2.1, we conclude

$$\Delta^s (D_0^+)^{2s} (\Delta^s R_{\mathcal{I}^+}^{[+s]}) = \mathfrak{C}_s^{(1)} R_{\mathcal{I}^+}^{[-s]}, \quad \mathfrak{C}_s^{(1)} := (2i\omega)^{2s}.$$

We now turn to $r = r_+$, and consider solutions with spin $-s$. We start with $|a| < M$; for any $s \neq 0$, using (2.16), we obtain

$$\begin{aligned} D_0^- R_{\mathcal{H}^+}^{[-s]} &= \left(\frac{d}{dr} - i \frac{\omega(r^2 + a^2)}{\Delta} + \frac{iam}{\Delta} \right) R^{[-s]} \\ &= \sum_{k=0}^{\infty} b_k^{[-s]} \left((\xi + k + s) - \frac{i\omega(r^2 + a^2)}{r - r_-} + \frac{iam}{r - r_-} \right) (r - r_+)^{\xi+k+s-1} \\ &= \sum_{k=0}^{\infty} b_k^{[-s]} \left((2\xi + k + s) - \frac{(\xi + i\omega(r + r_+))(r - r_+)}{r - r_-} \right) (r - r_+)^{\xi+k+s-1}. \end{aligned} \quad (2.19)$$

If $s \in \mathbb{Z}_{\geq 0}$, repeating the procedure $s-1$ more times yields

$$(D_0^-)^s R_{\mathcal{H}^+}^{[-s]} = \sum_{k=0}^{\infty} b_k^{[-s]} \left[\prod_{j=0}^{s-1} (2\xi + k + s - j) + O(r - r_+) \right] (r - r_+)^{\xi+k},$$

which is clearly regular at $r = r_+$ if $\omega = m\omega_+$; this regularity is preserved by acting with a further factor of $(D_0^-)^s$:

$$\Delta^s (D_0^-)^{2s} R_{\mathcal{H}^+}^{[-s]} = \Delta^s \left[s! f_- (b_0^{[-s]}, \dots, b_s^{[-s]}, \omega = m\omega_+, a, M) + O(r - r_+) \right],$$

for some function f_- which is independent of r . If, on the other hand, $s \in \frac{1}{2}\mathbb{Z}_{\geq 0} \setminus \mathbb{Z}$ or if $\omega \neq m\omega_+$, repeating the procedure of (2.19) $2s-1$ more times yields

$$\Delta^s (D_0^-)^{2s} R_{\mathcal{H}^+}^{[-s]} = \sum_{k=0}^{\infty} b_k^{[-s]} \left[\prod_{j=0}^{2s-1} (2\xi + k + s - j) + O(r - r_+) \right] \Delta^s (r - r_+)^{\xi+k-s}. \quad (2.20)$$

Finally, if $|a| = M$, then for any $s \neq 0$ and $\omega \neq m\omega_+$, using (2.17), we obtain

$$\begin{aligned} D_0^- R_{\mathcal{H}^+}^{[-s]} &= \left(\frac{d}{dr} - i\omega \frac{r^2 + M^2}{(r - M)^2} + \frac{iam}{(r - M)^2} \right) R^{[-s]} \\ &= \left(\frac{d}{dr} - \frac{\beta}{(r - M)^2} - \frac{(r + M)i\omega}{r - M} \right) e^{\beta(r-M)^{-1}} \sum_{k=0}^{\infty} b_k^{[-s]} (r - M)^{-2iM\omega+2s+k} \\ &= [-2\beta + O(r - M)] e^{\beta(r-M)^{-1}} \sum_{k=0}^{\infty} b_k^{[-s]} (r - M)^{-2iM\omega+2s-2+k}, \end{aligned}$$

which, repeating $2s - 1$ times, yields

$$\Delta^s(D_0^-)^{2s} R_{\mathcal{H}^+}^{[-s]} = \sum_{k=0}^{\infty} b_k^{[-s]} \left[(-2\beta)^{2s} + O(r-M) \right] e^{\beta(r-M)^{-1}} (r-M)^{-2iM\omega+k}. \quad (2.21)$$

By Lemma 2.5, $\Delta^s(D_0^-)^{2s} R_{\mathcal{H}^+}^{[-s]}$ satisfies the homogeneous spin $+s$ radial ODE (2.9), hence, by Definition 2.1, we conclude that

$$\Delta^s(D_0^-)^{2s} R_{\mathcal{H}^+}^{[-s]} = \mathfrak{C}_s^{(2)} \Delta^s R_{\mathcal{H}^+}^{[+s]},$$

where we have set

$$\mathfrak{C}_s^{(2)} := \begin{cases} \prod_{j=0}^{2s-1} (2\xi + s - j) & \text{if } |a| < M \text{ and either } \omega \neq m\omega_+ \text{ or } s \in \frac{1}{2}\mathbb{Z}_{\geq 0} \setminus \mathbb{Z}, \\ s! f_-(b_0^{[-s]}, \dots, b_s^{[-s]}, \omega, a, M), & \text{if } |a| < M, \omega = m\omega_+ \text{ and } s \in \mathbb{Z}_{\geq 0}, \\ (-2\beta)^{2s} & \text{if } |a| = M. \end{cases}$$

The cases $R_{\mathcal{H}^+}^{[+s]}$ and $R_{\mathcal{I}^+}^{[-s]}$. We need to take the same series-based approach as in the previous case. As we will see, for $R_{\mathcal{H}^+}^{[+s]}$ and $R_{\mathcal{I}^+}^{[-s]}$, the constant arising from application of the Teukolsky–Starobinsky operators to the series for these functions is more complicated to compute, as it involves up to $2s$ terms of their series expansion.

We begin with spin $-s$ at $r = \infty$. For any $s \geq \frac{1}{2}$, we have

$$\begin{aligned} D_0^- R_{\mathcal{I}^+}^{[-s]} &= \left(\frac{d}{dr} - i \frac{\omega(r^2 + a^2)}{\Delta} + \frac{iam}{\Delta} \right) \left[e^{i\omega r} r^{2iM\omega+2s-1} \sum_{k=0}^{2s} c_k^{[-s]} r^{-k} + O(r^{-2s-1}) \right] \\ &= e^{i\omega r} r^{2iM\omega+2s-1} \left[\sum_{k=0}^{2s} c_k^{[-s]} \left(\frac{2iM\omega(a^2 - 2Mr)}{r\Delta} + \frac{iam}{\Delta} + \frac{2s-1-k}{r} \right) r^{-k} \right]. \end{aligned} \quad (2.22)$$

The key difficulty, which is not present in the case of spin $+s$, is the cancellation that occurs between the ω dependence of D_0^- and that of the leading order behavior of $R_{\mathcal{I}^+}^{[-s]}$. For instance, if $s = \frac{1}{2}$, then

$$D_0^- R_{\mathcal{I}^+}^{[-1/2]} = \left[(-4iM^2\omega + iam) c_0^{[-1/2]} - c_1^{[-1/2]} + O(r^{-1}) \right] e^{i\omega r} r^{2iM\omega-2},$$

so taking D_0^- lowers the leading order behavior by a factor of r^{-2} . Indeed, an application of $(D_0^-)^{2s}$ lowers the leading order polynomial decay of $R_{\mathcal{I}^+}^{[-s]}$ by a factor of r^{-4s} and the coefficient of the leading order term on the right hand side depends on coefficients $c_0^{[-s]}, \dots, c_{2s}^{[-s]}$ of the asymptotic expansion for $R_{\mathcal{I}^+}^{[-s]}$:

$$\Delta^s(D_0^-)^{2s} R_{\mathcal{I}^+}^{[-s]} = \Delta^s e^{i\omega r} r^{2iM\omega-2s-1} \left[f_+ \left(c_0^{[-s]}, \dots, c_{2s}^{[-s]}, \omega, m, a, M \right) + O(r^{-1}) \right], \quad (2.23)$$

where f_+ is independent of r . By standard theory of asymptotic analysis, $c_1^{[-s]}, \dots, c_{2s}^{[-s]}$ are uniquely determined by $c_0^{[-s]}$ and the ODE parameters $(\omega, m, \lambda, s, a, M)$, so we can define

$$\mathfrak{C}_s^{(3)}(\omega, m, \lambda, s, a, M) := \frac{f_+ \left(c_0^{[-s]}, \dots, c_{2s}^{[-s]}, \omega, m, a, M \right)}{c_0^{[-s]}}.$$

By Lemma 2.5, (2.23) satisfies the spin $+s$ radial ODE. As before, by Definition 2.1, we conclude

$$\Delta^s(D_0^-)^{2s} R_{\mathcal{I}^+}^{[-s]} = \mathfrak{C}_s^{(3)} \Delta^s R_{\mathcal{I}^+}^{[+s]}.$$

Now consider spin $+s$ near $r = r_+$, and recall (2.16) and (2.17). If $|a| < M$, and $\omega \neq m\omega_+$ or $s \in \frac{1}{2}\mathbb{Z}_{\geq 0} \setminus \mathbb{Z}$, then

$$\begin{aligned} D_0^+ \left(\Delta^s R_{\mathcal{H}^+}^{[+s]} \right) &= \left(\frac{d}{dr} + \frac{i\omega(r+r_+)}{r-r_-} - \frac{\xi}{r-r_+} + \frac{\xi}{r-r_-} \right) \sum_{k=0}^{\infty} b_k^{[+s]} (r-r_+)^{\xi+k} (r-r_-)^s \\ &= \sum_{k=0}^{\infty} b_k^{[+s]} \left(\frac{k}{r-r_+} + \frac{\xi+s+i\omega(r+r_+)}{r-r_-} \right) (r-r_+)^{\xi+k} (r-r_-)^s, \end{aligned}$$

and if $|a| = M$,

$$\begin{aligned} D_0^+ \left(\Delta^s R_{\mathcal{H}^+}^{[+s]} \right) &= \left(\frac{d}{dr} + \frac{\beta}{(r-M)^2} + \frac{2Mi\omega}{r-M} + i\omega \right) \sum_{k=0}^{\infty} b_k^{[+s]} e^{\beta(r-M)^{-1}} (r-M)^{-2iM\omega+k} \\ &= \sum_{k=0}^{\infty} b_k^{[+s]} \left(\frac{k}{r-M} + i\omega \right) e^{\beta(r-M)^{-1}} (r-M)^{-2iM\omega+k}. \end{aligned}$$

As in the previous case, there is cancellation between the ξ or β dependence of the Teukolsky–Starobinsky operators and that of the leading order behavior of $\Delta^s R_{\mathcal{H}^+}^{[+s]}$. Proceeding as above, we note that application of $(D_0^+)^{2s}$ does not change the leading order polynomial decay of $\Delta^s R_{\mathcal{H}^+}^{[+s]}$ as $r \rightarrow r_+$ if $\omega \neq m\omega_+$ or, in case $|a| < M$, if $s \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}$, but makes the coefficient of the resulting leading order term depend on coefficients $b_0^{[+s]}, \dots, b_{2s}^{[+s]}$ of the asymptotic series for $R_{\mathcal{H}^+}^{[+s]}$. In the case $|a| < M$, $s \in \mathbb{Z}$ and $\omega = m\omega_+$, we have

$$\begin{aligned} D_0^+ \left(\Delta^s R_{\mathcal{H}^+}^{[+s]} \right) &= \left(\frac{d}{dr} + \frac{i\omega(r+r_+)}{r-r_-} \right) \sum_{k=0}^{\infty} b_k^{[+s]} (r-r_+)^k (r-r_-)^s \\ &= \sum_{k=0}^{\infty} b_k^{[+s]} [s+k+O(r-r_+)] (r-r_+)^{\xi+s+k-1} (r-r_-)^s \\ \implies (D_0^+)^s \left(\Delta^s R_{\mathcal{H}^+}^{[+s]} \right) &= \sum_{k=0}^{\infty} b_k^{[+s]} \left[\prod_{j=0}^{s-1} (s+k-j) + O(r-r_+) \right] (r-r_+)^k (r-r_-)^s \\ \implies \Delta^s (D_0^+)^{2s} \left(\Delta^s R_{\mathcal{H}^+}^{[+s]} \right) &= [s!g_-(b_0^{[-s]}, \dots, b_s^{[-s]}, \omega = m\omega_+, a, M) + O(r-r_+)] (r-r_+)^s, \end{aligned}$$

for some function g_- which is independent of r . Thus, we again appeal to theory of asymptotic analysis to define $\mathfrak{C}_s^{(4)}(\omega, m, \lambda, s, a, M)$ such that

$$\Delta^s (D_0^+)^{2s} \left(\Delta^s R_{\mathcal{H}^+}^{[+s]} \right) = \mathfrak{C}_s^{(4)} R_{\mathcal{H}^+}^{[-s]}.$$

The cases $R_{\mathcal{H}^-}^{[\pm s]}$ and $R_{\mathcal{I}^-}^{[\pm s]}$. Finally by direct inspection of (2.11)–(2.15), for $R_{\mathcal{H}^-}^{[\pm s]}$ and $R_{\mathcal{I}^-}^{[\pm s]}$, we can easily obtain the Teukolsky–Starobinsky identities by analogy with $R_{\mathcal{H}^+}^{[\mp s]}$ and $R_{\mathcal{I}^+}^{[\mp s]}$, respectively. $\square$

2.2 The angular case

2.2.1 The angular ODE and its solutions

Fix parameters $s \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, $m-s \in \mathbb{Z}$, $\nu \in \mathbb{R}$ and $\lambda \in \mathbb{R}$.

In this section, we discuss the angular ODE

$$-\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) \Xi^{[s]} + \left(\frac{(m+s \cos \theta)^2}{\sin^2 \theta} + \nu^2 \sin^2 \theta + 2\nu s \cos \theta \right) \Xi^{[s]} = (\lambda + \nu^2) \Xi^{[s]}, \quad (2.24)$$

where $\theta \in (0, \pi)$. It will be convenient to sometimes use the variable $x = \cos \theta$, so that (2.24) is considered for $x \in (-1, 1)$.

The ODE has regular singularities at $\theta = 0, \pi$ ($x = \pm 1$), and an irregular singularity at $\theta = \infty$. By standard asymptotic analysis, we can span the solution space by two linearly independent asymptotic solutions at each of the singularities $\theta = 0, \pi$. Our basis is given by the following:

Definition 2.6. Fix $s, m \in \frac{1}{2}\mathbb{Z}$ with $m - s \in \mathbb{Z}$, and $(\lambda, \nu) \in \mathbb{R}^2$.

1. Define $\Xi_{\mathcal{N}+}^{[s]}$ to be the unique classical solution to the angular ODE (2.24) such that

- i. $\Xi_{\mathcal{N}+}^{[s]}(\theta)(1 - \cos \theta)^{-\frac{1}{2}|m+s|}$ is smooth at $\theta = 0$,
- ii. $\left((1 - \cos \theta)^{-\frac{1}{2}|m+s|}\Xi_{\mathcal{N}+}^{[s]}\right)|_{\theta=0} = 1$.

2. Define $\Xi_{\mathcal{S}+}^{[s]}$ to be the unique classical solution to the angular ODE (2.24) such that

- i. $\Xi_{\mathcal{S}+}^{[s]}(\theta)(1 + \cos \theta)^{-\frac{1}{2}|m-s|}$ is smooth at $\theta = \pi$,
- ii. $\left((1 + \cos \theta)^{-\frac{1}{2}|m-s|}\Xi_{\mathcal{S}+}^{[s]}\right)|_{\theta=\pi} = 1$.

In light of the previous comments, we find that we have the following representation for solutions of (2.24):

Lemma 2.7. [TO FILL]

2.2.2 The angular Teukolsky–Starobinsky identities

In this section, we will introduce the Teukolsky–Starobinsky identities. We begin by defining the differential operators

$$L_n^\pm = \frac{d}{d\theta} \pm \left(\frac{m}{\sin \theta} - \nu \sin \theta \right) + n \cot \theta. \quad (2.25)$$

The angular Teukolsky–Starobinsky identities are differential identities that relate solutions of the angular ODE (2.24) with spin $+s$ and spin $-s$, and were introduced in [TP74, SC74] and extended for general s in [KMW89]. We will present a rigorous statement:

Proposition 2.8 (Angular Teukolsky–Starobinsky identities). Fix $s \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, $m - s \in \mathbb{Z}$ and $(\lambda, \nu) \in \mathbb{R}^2$. Consider regular solutions to the angular ODE (2.24), i.e. solutions that admit a representation

$$\begin{aligned} \Xi^{[\pm s]} &= a_{\mathcal{N}+}^{[\pm s]} \Xi_{\mathcal{N}+}^{[\pm s]} \\ &= a_{\mathcal{S}+}^{[\pm s]} \Xi_{\mathcal{S}+}^{[\pm s]}, \end{aligned}$$

for some complex constants $a_{\mathcal{N}+}^{[\pm s]}, a_{\mathcal{S}+}^{[\pm s]}$. Then we have

$$\begin{aligned} (\sin \theta)^{2s} \left(\frac{L_s^+}{\sin \theta} \right)^{2s} \Xi^{[+s]} &= \mathfrak{B}_s^{(1)} a_{\mathcal{N}+}^{[+s]} \Xi_{\mathcal{N}+}^{[-s]} \\ &= \mathfrak{B}_s^{(4)} a_{\mathcal{S}+}^{[+s]} \Xi_{\mathcal{S}+}^{[-s]}, \\ (\sin \theta)^{2s} \left(\frac{L_s^-}{\sin \theta} \right)^{2s} \Xi^{[-s]} &= \mathfrak{B}_s^{(3)} a_{\mathcal{N}+}^{[-s]} \Xi_{\mathcal{N}+}^{[+s]} \\ &= \mathfrak{B}_s^{(2)} a_{\mathcal{S}+}^{[-s]} \Xi_{\mathcal{S}+}^{[+s]}, \end{aligned} \quad (2.26)$$

for some $\mathfrak{B}_s^{(i)} \in \mathbb{C} \setminus \{0\}$, $i = 1, \dots, 4$, which depend only on s , m and (λ, ν) and such that $\mathfrak{B}_0^{(i)} = 1$ for $s = 0$ and, for $s \neq 0$,

$$\begin{aligned}\mathfrak{B}_s^{(1)} &= \dots, \\ \mathfrak{B}_s^{(2)} &= (-1)^{2s} \mathfrak{B}_s^{(1)}, \\ \mathfrak{B}_s^{(3)} &= \dots, \\ \mathfrak{B}_s^{(4)} &= (-1)^{2s} \mathfrak{B}_s^{(3)}.\end{aligned}$$

In order to prove Proposition 2.8, we recall the following two results from [KMW89]:

Lemma 2.9. *For solutions of the angular ODE (2.24) with spin $\pm s$, we have*

$$\left[L_{1-s}^\mp L_s^\pm \mp 2(2s-1)\nu \cos \theta \right] \Xi^{[\pm s]} = -(\lambda + \nu^2 - 2m\nu + |s|) \Xi^{[\pm s]} \quad (2.27)$$

By induction on $2s$, one can also show that:

Lemma 2.10. *The operators $L_s^\pm$, defined in (2.25), satisfy the commutation relation*

$$\begin{aligned}(\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \left[L_{1-s}^\mp L_s^\pm \mp 2(2s-1)\nu \cos \theta \right] \\ = \left[L_{1-s}^\pm L_s^\mp \pm 2(2s-1)\nu \cos \theta \right] (\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s}.\end{aligned} \quad (2.28)$$

Combining the two lemmas, we see that, if $\Xi^{[\pm s]}$ solves (2.27) for spin $\pm s$, then $(\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \Xi^{[\pm s]}$ is a solution to the angular ODE (2.27) with spin $\mp s$. We are now ready to prove Proposition 2.8:

Proof of Proposition 2.8. The statement is trivial for $s = 0$. For $s \neq 0$, we obtain the result by using the asymptotic representations of $\Xi_{\mathcal{N}^+}$ and $\Xi_{\mathcal{S}^+}$ with respect to the variable $x = \cos \theta$,

$$\Xi_{\mathcal{N}^+}^{[\pm s]} = \sum_{k=0}^{\infty} b_k^{[\pm s]} (1-x)^{\frac{1}{2}|m \pm s| + k}, \quad \Xi_{\mathcal{S}^+}^{[\pm s]} = \sum_{k=0}^{\infty} c_k^{[\pm s]} (1+x)^{\frac{1}{2}|m \mp s| + k},$$

on which we can act directly with the Teukolsky–Starobinsky operators. By the normalizations, $c_0^{[\pm s]} = b_0^{[\pm s]} = 1$. Note also that

$$\frac{L_s^\pm}{\sin \theta} = -\frac{d}{dx} + \frac{sx \pm m}{1-x^2} \mp \nu = -\frac{d}{dx} + \frac{s \pm m}{2(1-x)} - \frac{s \mp m}{2(1+x)} \mp \nu. \quad (2.29)$$

The $\theta = 0$ end. We begin by computing the effect of applying the Teukolsky–Starobinsky operators to $\Xi_{\mathcal{N}^+}^{[\pm s]}$. In doing so, it will be useful to note that, from (2.29),

$$\left(\frac{L_s^\pm}{\sin \theta} \right) \left((1-x)^{-\frac{1}{2}(s \pm m)} \Xi \right) = (1-x)^{-\frac{1}{2}(s \pm m)} \left(-\frac{d}{dx} - \frac{s \mp m}{2(1+x)} \mp \nu \right) \Xi,$$

for any function $\Xi(x)$. It follows immediately that, if $s \pm m \leq 0$, then

$$\begin{aligned}(\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \Xi_{\mathcal{N}^+}^{[\pm s]} &= (\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \left[(1-x)^{-\frac{1}{2}(s \pm m)} \sum_{k=0}^{\infty} b_k^{[\pm s]} (1-x)^k \right] \\ &= (1+x)^s (1-x)^{\frac{1}{2}|-s \pm m|} \left(-\frac{d}{dx} - \frac{s \mp m}{2(1+x)} \mp \nu \right)^{2s} \sum_{k=0}^{\infty} b_k^{[\pm s]} (1-x)^k.\end{aligned}$$

Combining Lemmas 2.10, 2.9 and Definition 2.6, we deduce that $(\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \Xi_{\mathcal{N}^+}^{[\pm s]}$ is a solution to the $\mp s$ angular ODE with no irregular behavior at $\theta = 0$. Thus, there are functions $f_\pm$ so that

$$\begin{aligned} s + m \leq 0 &\implies \mathfrak{B}_s^{(1)} = 2^s \left((2s)! b_{2s}^{[+s]} + f_+(b_0^{[+s]}, \dots, b_{2s-1}^{[+s]}) \right), \\ s - m \leq 0 &\implies \mathfrak{B}_s^{(3)} = 2^s \left((2s)! b_{2s}^{[-s]} + f_-(b_0^{[-s]}, \dots, b_{2s-1}^{[-s]}) \right). \end{aligned}$$

If $s \pm m \geq 1$, then

$$\begin{aligned} \left(\frac{L_s^\pm}{\sin \theta} \right) \Xi_{\mathcal{N}^+}^{[\pm s]} &= \sum_{k=0}^{\infty} b_k^{[\pm s]} [k + s \pm m + O(1-x)] (1-x)^{\frac{1}{2}(s \pm m - 2) + k} \\ \implies \left(\frac{L_s^\pm}{\sin \theta} \right)^n \Xi_{\mathcal{N}^+}^{[\pm s]} &= \sum_{k=0}^{\infty} b_k^{[\pm s]} \left[\prod_{j=0}^{n-1} (k + s \pm m - j) + O(1-x) \right] (1-x)^{\frac{1}{2}(s \pm m - 2n) + k}. \end{aligned}$$

It follows that:

- if $s \pm m \geq 2s \geq 1$, then

$$\begin{aligned} (\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \Xi_{\mathcal{N}^+}^{[\pm s]} &= (1-x^2)^s \sum_{k=0}^{\infty} b_k^{[\pm s]} \left[\prod_{j=0}^{2s-1} (k + s \pm m - j) + O(1-x) \right] (1-x)^{\frac{1}{2}(-s \pm m) - s + k} \\ &= \sum_{k=0}^{\infty} b_k^{[\pm s]} \left[2^s \prod_{j=0}^{2s-1} (k + s \pm m - j) + O(1-x) \right] (1-x)^{\frac{1}{2}|-s \pm m| + k}, \end{aligned}$$

so, combining Lemmas 2.10, 2.9 and Definition 2.6, we deduce

$$\begin{aligned} s + m \geq 2s \geq 1 &\implies \mathfrak{B}_s^{(1)} = 2^s \prod_{j=0}^{2s-1} (s + m - j), \\ s - m \geq 2s \geq 1 &\implies \mathfrak{B}_s^{(3)} = 2^s \prod_{j=0}^{2s-1} (s - m - j); \end{aligned}$$

- otherwise, if $1 \leq s \pm m \leq 2s - 1$, then

$$\begin{aligned} (\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \Xi_{\mathcal{N}^+}^{[\pm s]} &= (\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{s \mp m} \left(\frac{L_s^\pm}{\sin \theta} \right)^{s \pm m} \Xi_{\mathcal{N}^+}^{[\pm s]} \\ &= (\sin \theta)^{2s} \left(\frac{L_s^\pm}{\sin \theta} \right)^{s \mp m} \left[(1-x)^{-\frac{1}{2}(s \pm m)} \sum_{k=0}^{\infty} b_k^{[\pm s]} \left[\frac{(k + s \pm m)!}{k!} + O(1-x) \right] (1-x)^k \right] \\ &= (1+x)^s (1-x)^{\frac{1}{2}|-s \pm m|} \times \\ &\quad \times \left(-\frac{d}{dx} - \frac{s \mp m}{2(1+x)} \mp \nu \right)^{s \mp m} \left[\sum_{k=0}^{\infty} b_k^{[\pm s]} \left[\frac{(k + s \pm m)!}{k!} + O(1-x) \right] (1-x)^k \right], \end{aligned}$$

so, combining Lemmas 2.10, 2.9 and Definition 2.6, we deduce there are functions $g_\pm$ so that

$$\begin{aligned} 1 \leq s + m \leq 2s - 1 &\implies \mathfrak{B}_s^{(1)} = 2^s (2s)! \left(b_{s-m}^{[+s]} + g_+(b_0^{[+s]}, \dots, b_{s-m-1}^{[+s]}) \right), \\ 1 \leq s - m \leq 2s - 1 &\implies \mathfrak{B}_s^{(3)} = 2^s (2s)! \left(b_{s+m}^{[-s]} + g_-(b_0^{[-s]}, \dots, b_{s+m-1}^{[-s]}) \right). \end{aligned}$$

The $\theta = \pi$ end. To compute $\mathfrak{B}_s^{(2)}$ and $\mathfrak{B}_s^{(4)}$, we can follow the same steps as before. Alternatively, we can appeal to the symmetry of (2.24) under $\theta \mapsto \pi - \theta$: $L_s^\pm \mapsto -L_s^\mp$ and $\Xi_{\mathcal{N}^+}^{[\pm s]}(\theta) = \Xi_{\mathcal{N}^+}^{[\mp s]}(\pi - \theta)$, so the claimed identities follow easily.

Non-vanishing of the constants. Finally, we turn to the proof that $\mathfrak{B}_s^{(i)}$ are non-zero. In view of the relationships between them, it is enough to show $\mathfrak{B}_s^{(1)}$ and $\mathfrak{B}_s^{(3)}$ are nonzero. Note that, from (2.29),

$$\frac{L_s^\pm}{\sin \theta} \left(e^{\mp \nu x} (1-x)^{-\frac{1}{2}(s \pm m)} (1+x)^{-\frac{1}{2}(s \mp m)} \right) = 0.$$

Thus, if we write $\Xi^{[\pm s]} = e^{\mp \nu x} (1-x)^{-\frac{1}{2}(s \pm m)} (1+x)^{-\frac{1}{2}(s \mp m)} P^{[\pm s]}(x)$ for some $P^{[\pm s]}: (-1, 1) \rightarrow \mathbb{C}$, we obtain

$$\left(\frac{L_s^\pm}{\sin \theta} \right)^{2s} \Xi^{[\pm s]} = (-1)^{2s} e^{\mp \nu x} (1-x)^{-\frac{1}{2}(s \pm m)} (1+x)^{-\frac{1}{2}(s \mp m)} \frac{d^{2s}}{dx^{2s}} P_s^{[\pm s]}(x).$$

We have established that so far that, if $\Xi^{[\pm s]}$ solves the angular ODE (2.24) and is regular at $\theta = 0, \pi$, then it satisfies the identities in the statement for some $\mathfrak{B}_s^{(1)}$ and $\mathfrak{B}_s^{(3)}$, respectively. From the above, it follows that: if $\mathfrak{B}_s^{(1)} = 0$, for regular solutions $\Xi^{[+s]}$, $P^{[+s]}$ is a polynomial of degree at most $2s - 1$; if $\mathfrak{B}_s^{(3)} = 0$, for regular solutions $\Xi^{[-s]}$, then $P^{[-s]}$ must be a polynomial of degree at most $2s - 1$. However, the regularity of $\Xi^{[\pm s]}$ forces $P^{[\pm s]}$ to have at least $2s$ zeros (counted with multiplicity), so $\mathfrak{B}_s^{(i)} \neq 0$. Indeed:

- regularity at $\theta = 0$ implies $P^{[\pm s]}$ must vanish to order at least $\frac{1}{2}|s \pm m| + \frac{1}{2}(s \pm m) = \max\{s \pm m, 0\}$ at $x = 1$;
- regularity at $\theta = \pi$ implies $P^{[\pm s]}$ must vanish to order at least $\frac{1}{2}|s \mp m| + \frac{1}{2}(s \mp m) = \max\{s \mp m, 0\}$ at $x = -1$;

and thus the polynomial $P^{[\pm s]}$ must have at least $\max\{s \pm m, 0\} + \max\{s \mp m, 0\} \geq 2s$ roots on the real line. $\square$